



IPD GROUP 38

AIR VISION

Predicting tomorrow's AQI with AI

AADITYA MEHTA

ANSH SABOO

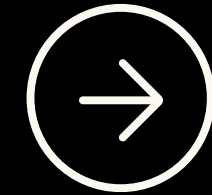
ARYAN SANGHANI

HARDIK IYER



PROF ANIKET KORE
GROUP MENTOR

VISION FOR THIS SEMESTER



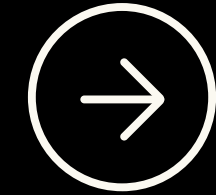
In the last semesters,

- Formulated a mathematical model to compute garbage collected from different wards of Mumbai reaching the Deonar dumping ground

In this semester,

- Developed a user-oriented website with focus on accessibility and ease of use.
- Sought guidance from mentor to improve accuracy and reliability of calculations
- Studied and compared two forecasting models: SARIMA and Prophet
- Evaluated performance and suitability of both models for the dataset
- Selected Prophet as the more appropriate model for project requirements
- Generated future index predictions using Prophet
- Integrated predictions into the formula for better precision
- Achieved a data-driven approach to analyze and project waste management patterns

MOTIVATION FOR THE IDEA



A critical gap in existing research and data has been identified:
the impact of landfill incinerations on AQI.

- Solving discrepancies in already existing datasets: Inefficiencies and inaccuracies in current waste management practices exacerbate pollution levels.
- Helping government bodies to drive Policy and Action: A data-driven approach to predicting AQI spikes caused by landfill fires

OUR APPLICATION



Environmental Impact Analysis Platform

Sign In

Access your AirVision dashboard

Email

Password



Select Access Type:

 User Dashboard

 Admin Panel

Our application has two panels: User and Admin.

- **User Panel:** The user only inputs the ward name, while other parameters like population density, education index, and urbanization index are auto-fetched from the backend.
- **Admin Panel:** Designed for professionals, it allows manual entry and updating of all parameters for greater flexibility and accuracy.

This setup ensures ease of use for general users while giving full control to experts.



Mumbai Waste Calculator

Calculate waste generation patterns across Mumbai's administrative wards and understand their environmental impact.

Location Selection

Select your Mumbai location to calculate waste generation data

Mumbai Location

Select location...

Calculation Parameters

Daily waste per person: 0.3 kg Festival surge factor: 20%
Monsoon surge factor: 10% Annual calculation: 365 days

Calculation Results

Annual waste generation data for selected location



Select a location to view results

Calculation Formula

Mathematical model for waste generation estimation

$$\text{Total Waste} = P \times Wp \times Df \times Ef \times Uf \times (1 + Sf + Sm) \times 365$$

P = Population (inhabitants)

Df = Population density factor

Uf = Built-up area factor

Sm = Monsoon surge factor (0.1)

Wp = Daily waste per person (0.3 kg)

Ef = Education index factor

Sf = Festival surge factor (0.2)

365 = Annual calculation period

[← Back to Login](#)

[ADMIN PANEL](#)



Waste Generation Calculator

Calculate annual waste generation using population density, education index, and built-up area parameters.

Input Parameters

Enter the required data for waste calculation

Population (P)

Number of people in the area

Population Density (persons/km²)

Population per square kilometer

Education Index (0-1)

A composite score reflecting the quality and outcomes of municipal education at the ward level

Built-up Area (%)

The total space covered by buildings and structures, including usable area, walls, and attached balconies

[Calculate Waste Generation](#)

Calculation Results

Annual waste generation analysis



Enter parameters to calculate results

COMPARISON BETWEEN PROPHET AND SARIMA

PROPHET

- Additive model developed by Facebook
- Works with non-stationary data
- Minimal tuning, easy to implement
- Handles multiple seasonalities (daily, weekly, yearly)
- Robust to missing data and outliers
- Provides interpretable components (trend, seasonality, holidays)
- Best suited for large, noisy, real-world datasets

SARIMA

- Traditional statistical time-series model
- Requires data to be stationary
- Needs manual tuning of parameters (p, d, q, P, D, Q, m)
- Handles single seasonality well
- High accuracy if parameters are optimized
- Sensitive to missing data and outliers
- Best suited for structured, smaller datasets

WHY WE CHOSE PROPHET

We compared SARIMA and Prophet for our forecasting task.

SARIMA, while powerful, requires data to be stationary and involves complex parameter tuning. It also does not easily allow the inclusion of external factors.

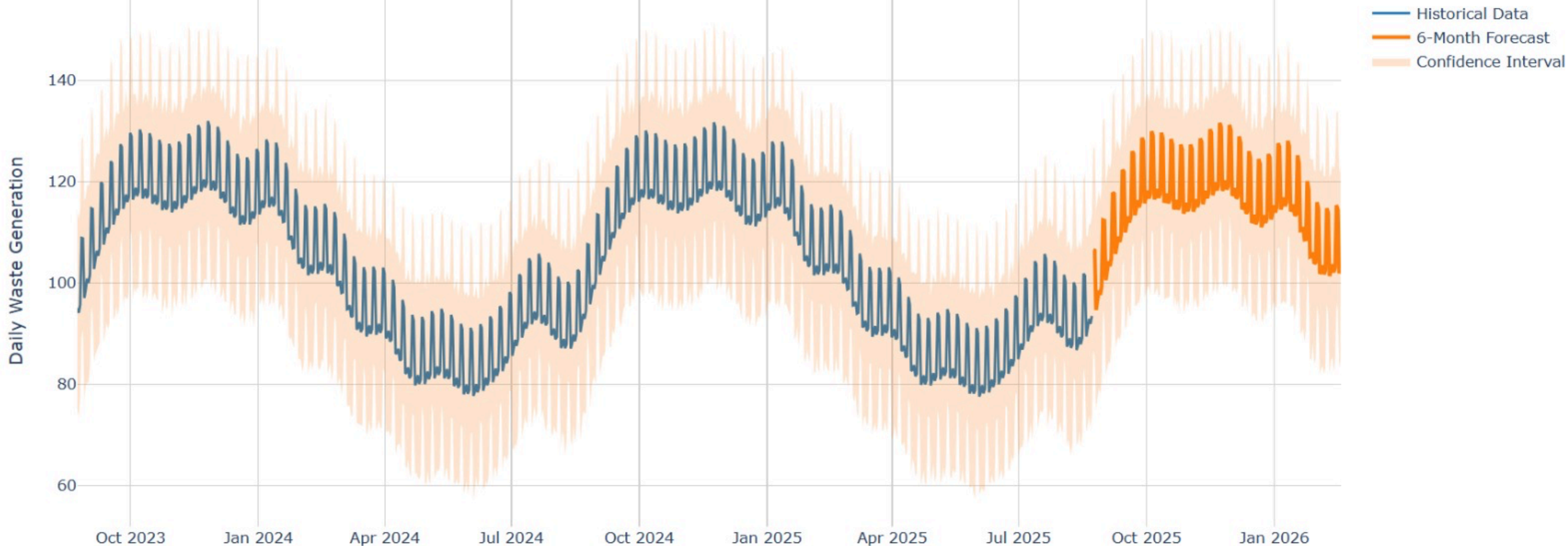
Prophet, on the other hand, handles non-stationary and messy data more effectively.

It allows us to add key regressors such as population density, education index, urbanization, monsoon, and festivals.

This made Prophet the more suitable choice for building accurate and adaptable waste generation forecasts.

Waste Generation Forecast - Andheri West/Jogeshwari West/Vile Parle(west)/Juhu

Andheri West/Jogeshwari West/Vile Parle(west)/Juhu ▼





AIR VISION

THANK YOU

for your time and attention

Presented by IPD GROUP 38

